Strategic Churn Mitigation

Transforming At-Risk Revenue into Lifecycle Value

PREPARED FOR EXECUTIVE BOARD | 2026


$34.6 Million
Annual revenue lost to preventable churn
49.3% of total revenue at risk
~50% of customers churn annually
$58.21 avg monthly revenue per lost customer

TELECOM CHURN: A GLOBAL CRISIS

20-40% annual churn = telecom industry average

Source: PLoS ONE 2022

Acquiring new customer is **5-10x costlier** than retaining existing

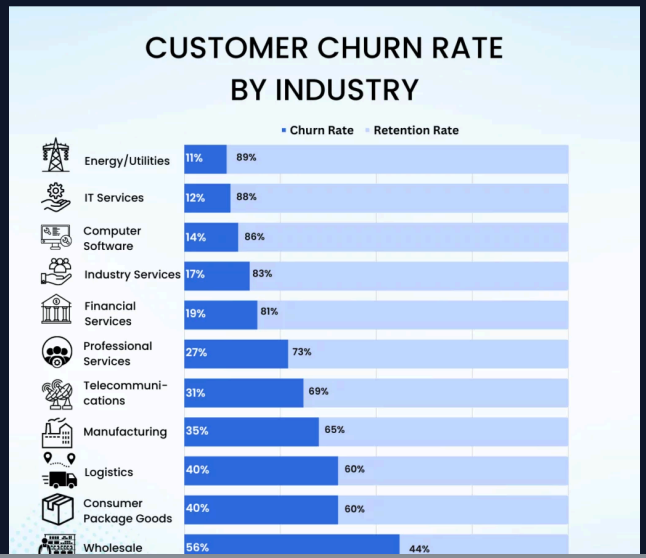
Source: PLoS ONE 2022

AI/ML intervention reduces churn by **up to 15%**

Source: McKinsey 2024 via Tridens

Personalized retention yields **15% loyalty increase**

Source: Tridens Technology 2024



COMPANY A'S RETENTION CRISIS IS STRUCTURAL

1. The Scale

49.9%

Churn rate — placing the company at or above industry worst. This is an existential threat to long-term profitability.

2. The Timing

69%

Of churners were customers for 12+ months. Peak churn window hits exactly at months 12-17 (55% churn rate).

3. The Silence

1.60 vs 1.98

Churning customers show NO increase in complaints (CustCare calls lower for churners). They leave without warning.

THE CUSTOMER JOURNEY

ENGAGED → DECLINING → SILENT → GONE

THREE AT-RISK PROFILES IDENTIFIED

Segment A: Silent Disengager

- **69%** of all churners
- Usage drops **-22%** before leaving
- Never escalate to support

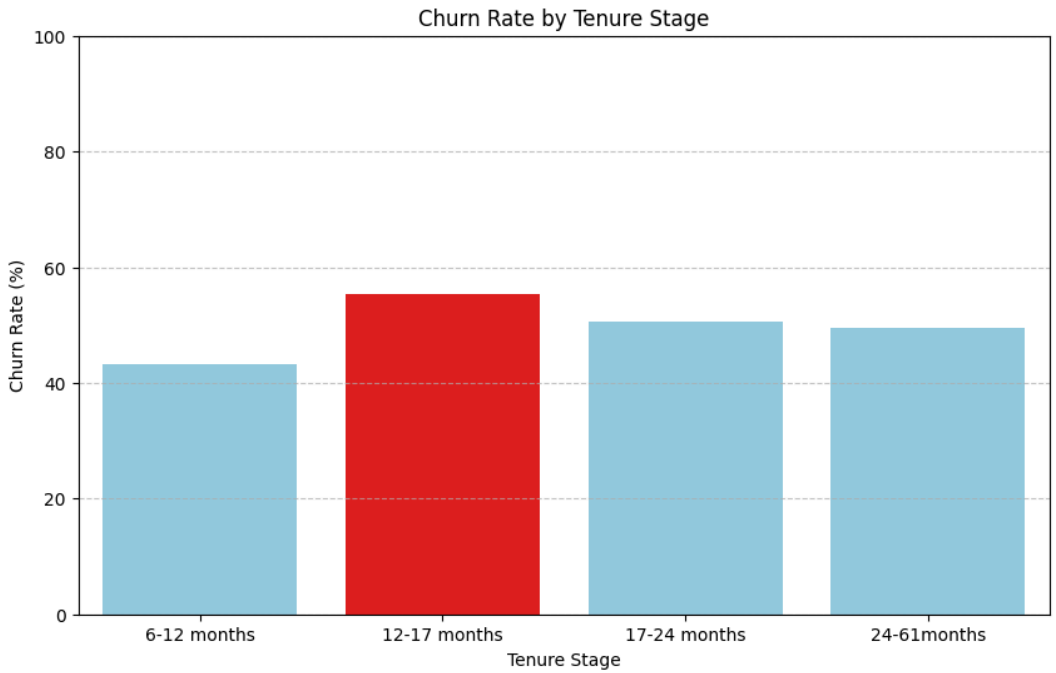
Segment B: Multi-Sub Household

- 4+ lines on account
- **55-78%** churn rate
- Household switch = multiplied loss

Segment C: Roaming User

- **55.9%** churn vs 44.1% retained
- Plan mismatch drives anxiety
- Bill shock triggers cancellation

THE RENEWAL WINDOW: WHERE CUSTOMERS DECIDE TO LEAVE



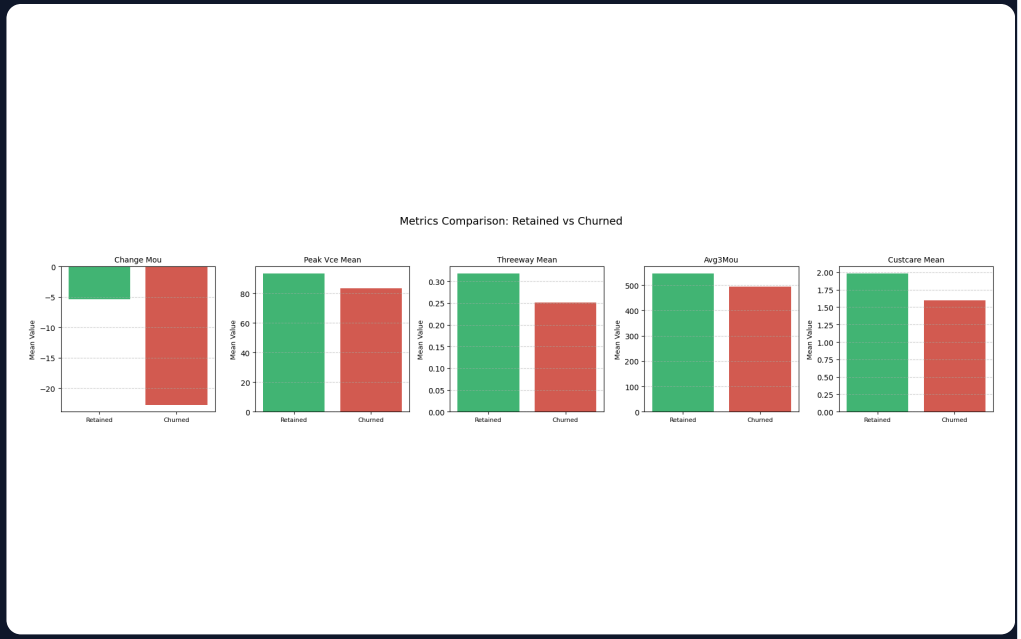
"55% of customers in their 12-17 month window will churn — the exact contract renewal decision point. This is not an onboarding failure. Company A is losing customers it already won."

SILENT DISENGAGEMENT: THE INVISIBLE CHURN SIGNAL

The Cascade

- Step 1: Usage begins declining (MOU drops -22%)
- Step 2: Call frequency drops (-9%)
- Step 3: Feature usage falls (Threeway calls -21%)
- Step 4: No complaint filed with Care
- Step 5: Renewal window hit → EXIT

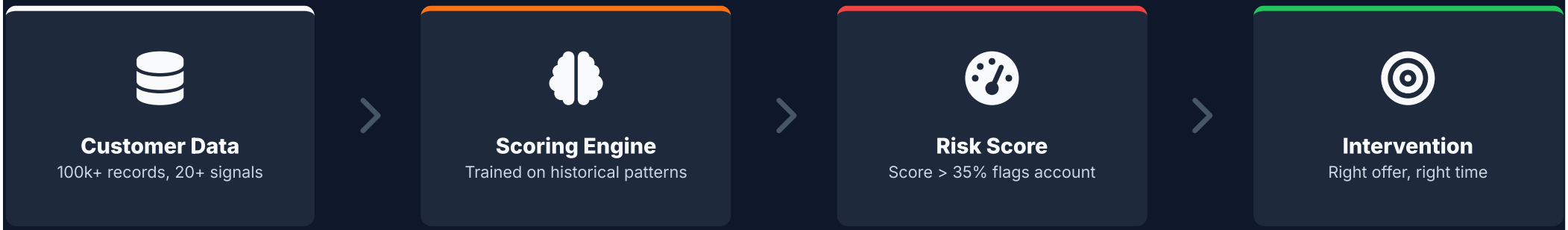
"By the time a user cancels, the decision was made weeks or months earlier. Usage decline over 30-60 days is the strongest early churn signal."



Revenue gap = only -1.7% | Usage gap = -11%
Churners are paying similar amounts but feel they get significantly less value.

BUILDING AN EARLY WARNING RADAR FOR CHURN

"Just as weather radar predicts storms before they hit, **ChurnGuard** detects churn signals before customers actively decide to leave."



MODEL SELECTION: CHAMPIONING ACCURACY AT SCALE

MODEL	AUC-ROC	STATUS
Logistic Regression	0.625	Baseline
CatBoost	0.672	Evaluated
XGBoost	0.672	Evaluated
LightBGM	**0.673**	**★ FINAL**

SELECTED ENGINE

METRIC (AUC-ROC)

0.673

THRESHOLD TARGET

0.35 - 0.38

RECALL (CAPTURE RATE)

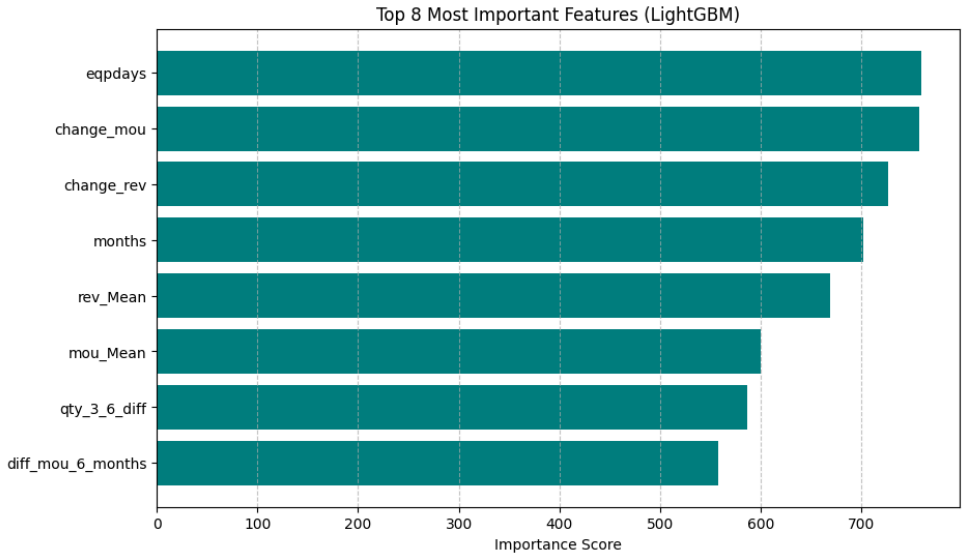
~91%

PRECISION

~55%

"At chosen sensitivity: 9 in 10 at-risk customers are correctly identified. Over 55% of flagged customers are genuine flight risks. Better than baseline random guessing by 34.6%."

WHAT DRIVES CHURN? MODEL AND DATA AGREE.



TIMING IS EVERYTHING

Device age + tenure = top predictors. Churn peaks at contract renewal, not randomly.

USAGE TELLS THE STORY

Declining minutes of use is the strongest behavioral signal, visible months before actual churn.

VALUE PERCEPTION BREAKS DOWN

Customers paying more per minute of use are 40%+ more likely to churn. Wrong plan = silent dissatisfaction.

CHURNGUARD: THREE INTERVENTIONS, ONE SYSTEM

Silent Disengager

69% of churners

Trigger: MOU drops >15% + months 10-17

Action: Personalized usage insight report + proactive plan optimization offer + device upgrade bundle.



App + SMS 60 days pre-renewal

Multi-Sub Household

High flight risk group

Trigger: 4+ lines + any risk signal

Action: Dedicated account manager assigned + customized household loyalty bundle offer.



Direct Account Outreach Call

Roaming User

Bill shock sensitive

Trigger: Roaming ratio > 0 + overage detected

Action: Real-time bill alert mid-cycle + automatic one-click roaming plan upgrade offer.



Real-time SMS during event

THE BUSINESS CASE: $12.5M NET RETURN

*Assumptions: 30% retention rate when intervened | $15 avg intervention cost | 150k user projection

REVENUE AT RISK

$34.6M

INTERVENTION COST

$1.85M

(Flags 123k customers)

REVENUE RECOVERED

$14.3M

(91% caught x 30% saved)

NET GAIN: $12,455,600

7.63x ROI — Every $1 spent yields $7.63 returned.

FROM POC TO PRODUCTION: 90-DAY ROADMAP

PHASE 1: Foundation

Days 1-30

- [ ] Deploy scoring engine to full base
- [ ] Identify immediate 12-17m risks
- [ ] Train retention team on playbooks

KPI: First risk scores generated

PHASE 2: Launch

Days 31-60

- [ ] Seg A: Automated usage reports live
- [ ] Seg B: Account manager assignments
- [ ] A/B test: Intervened vs Control

KPI: First retention lift measured

PHASE 3: Optimize

Days 61-90

- [ ] Measure actual vs projected ROI
- [ ] Retrain model with new labels
- [ ] Expand device refresh program

KPI: Full rollout decision secured

Month 4+: Full Deployment. Model retrains monthly. Estimated Breakeven: Day 45.

LIMITATIONS & RISK MITIGATION

IDENTIFIED RISK	MITIGATION STRATEGY
AUC 0.673 = imperfect predictor False positives will occur.	**A/B test before full rollout.** Monthly model retraining improves accuracy over time.
No competitor pricing data included Blind spot to external offers.	External market signals and pricing scrape data will be integrated into the **v2 model pipeline.**
Income variable data quality unclear Demographic data is sparse.	Excluded from final model scoring to prevent bias. **Demographic enrichment recommended** for future.
Retention offers visible to loyal customers Risk of margin cannibalization.	Target **ONLY** model-flagged segments via private channels. Strict policy against mass retention campaigns.

REFERENCES & METHODOLOGY

[1] PLoS ONE (2022) — *Churn prediction in telecom*

[2] Tridens Technology (2024) — *Telecom churn causes and prevention*

[3] Scalence (2026) — *Predictive analytics for churn reduction*

[4] Atticus Li (2026) — *Behavioral churn signals in subscription models*

[5] Databricks (2026) — *The Telecom Churn Intervention Window*

[6] Action Edge (2026) — *Predictive research in modern telecom*

[7] Vibetrace.com - Average churn per industry graph

[AI] Claude.ai (https://claude.ai/share/e589505a-ed63-4832-b887-b4d5eff42eef)